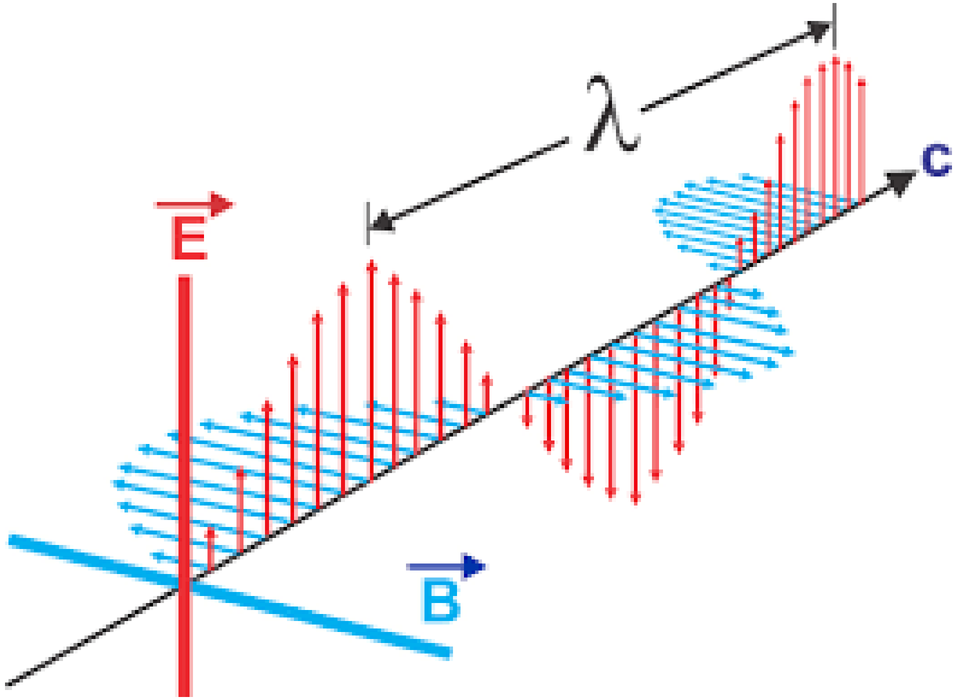
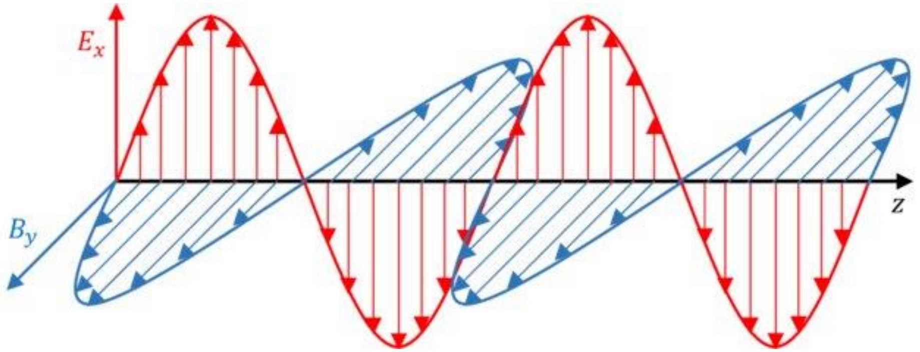
Electromagnetic Wave
Part-2





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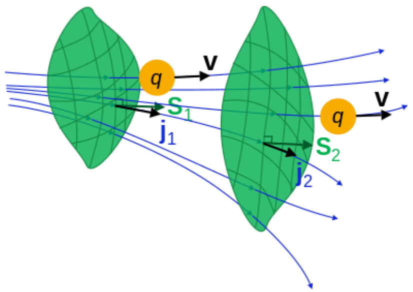
Electromagnetic wave in conductor

Electromagnetic Waves in Conductors



In the case of conductors, $\overrightarrow{J_f}$ is not zero.

The current density in a conductor is proportional to the electric field.

$$\overrightarrow{J_f} = \sigma\vec{E} \quad \text{.... (1)} \qquad \text{(Ohm's law)}$$

With this, Maxwell's equation for linear conducting media assume the form

$$(i)\ \vec{\nabla}.\vec{E} = \frac{\rho_f}{\epsilon}$$

$$(ii)\ \vec{\nabla}.\vec{B}=0$$

$$(iii)\ \vec{\nabla}X\vec{E}=-\frac{\partial\vec{B}}{\partial t}$$

$$(iv)\ \vec{\nabla}X\vec{B}=\mu\sigma\vec{E}+\mu\epsilon\frac{\partial\vec{E}}{\partial t} \quad \text{.................... (2)}$$

$$\text{(Rate that } q \text{ is flowing through the imaginary surface } S) = \iint_S \mathbf{j}\cdot d\mathbf{S}$$

$$\frac{dq}{dt} + \oiint_S \mathbf{j}\cdot d\mathbf{S} = \Sigma$$

Σ is the net rate that *q* is being generated inside the volume *V* per unit time

The continuity equation for free charge,

$$\vec{\nabla}.\overrightarrow{J_f} = -\frac{\partial\,\rho_f}{\partial t} \quad \text{...................... (3)}$$

Using equation (1) in (3), we get

$$\sigma(\vec{\nabla}.\vec{E}\ ) = -\frac{\partial\,\rho_f}{\partial t}$$

But $\vec{\nabla}.\vec{E} = \dfrac{\rho_f}{\epsilon}$

Therefore, $\quad \sigma(\frac{\rho_f}{\epsilon}) = -\frac{\partial\,\rho_f}{\partial t}$

Or, $\quad \rho_f(\frac{\sigma}{\epsilon}) + \frac{\partial\,\rho_f}{\partial t} = 0 \quad$ .... (4)

So, $\quad \rho_f(t) = \rho_f(0)\, e^{-(\sigma/\epsilon)t}$ .... (5)

$\Rightarrow$ Any initial free charge $\rho_f(0)$ dissipates in a characteristics time $\tau = {}^{\epsilon}/_{\sigma}$.

It means that if we put some free charge on a conductor, it will flow out to the edges.

The time constant τ is a measure of how "good" a conductor is.

- For a good conductor, τ is much less than the other relevant times in the problem (e.g., in oscillatory system, this means $\tau \ll \frac{1}{\omega}$)
- For a "poor" conductor, τ is greater than the characteristics times in the problem ($\tau \gg \frac{1}{\omega}$)

Here we are not interested in this transient behaviour, we will wait for any accumulated free charge to disappear .Then, $\overrightarrow{\rho_f} = 0$.

Then, we will have following Maxwell's equation

$$(i)\vec{\nabla}.\vec{E} = 0$$

$$(ii)\vec{\nabla}.\vec{B}=0$$

$$(iii)\vec{\nabla}X\vec{E}=-\frac{\partial\vec{B}}{\partial t}$$

$$(iv)\ \vec{\nabla}X\vec{B}=\mu\sigma\vec{E}+\mu\epsilon\frac{\partial\vec{E}}{\partial t}$$

.... (6)

Now, applying curl to equations (iii) and (iv):

Equation (iii) is:

$$\vec{\nabla}\mathrm{X}(\vec{\nabla}\mathrm{X}\vec{\mathrm{E}}) = -\frac{\partial(\vec{\nabla}\mathrm{X}\vec{\mathrm{B}})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E} = -\frac{\partial}{\partial t}\left(\mu\sigma\vec{E}+\mu\epsilon\frac{\partial\vec{E}}{\partial t}\right)$$

Or, $-\nabla^2\vec{E} = -\mu\epsilon\frac{\partial^2\vec{E}}{\partial t^2} - \mu\sigma\frac{\partial\vec{E}}{\partial t}$

$$\nabla^2\vec{E} = \mu\epsilon\frac{\partial^2\vec{E}}{\partial t^2} + \mu\sigma\frac{\partial\vec{E}}{\partial t} \quad \text{.....................} \ (7)$$

Equation (iv) is :

$$\vec{\nabla}\mathrm{X}(\vec{\nabla}\mathrm{X}\vec{\mathrm{B}}) = \mu\sigma\overrightarrow{(\vec{\nabla}XE)}+\mu\epsilon\frac{\partial(\vec{\nabla}X\vec{E})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla}.\vec{B}) - \nabla^2\vec{B} = -\mu\epsilon\frac{\partial^2\vec{B}}{\partial t^2} - \mu\sigma\frac{\partial\vec{B}}{\partial t}$$

$$\nabla^2\vec{B} = \mu\epsilon\frac{\partial^2\vec{B}}{\partial t^2} + \mu\sigma\frac{\partial\vec{B}}{\partial t} \quad \text{......................} \ (8)$$

Thus, we will have

$$\boldsymbol{\nabla^2\vec{E} = \mu\epsilon\frac{\partial^2\vec{E}}{\partial t^2} + \mu\sigma\frac{\partial\vec{E}}{\partial t}}$$

$$\boldsymbol{\nabla^2\vec{B} = \mu\epsilon\frac{\partial^2\vec{B}}{\partial t^2} + \mu\sigma\frac{\partial\vec{B}}{\partial t}}$$

Note: 3D wave equations for E and B in a conductor have an additional term that has a single time derivative – which is analogous to a velocity dependent damping term associated with the motion of a 1D mechanical harmonic oscillator.

The general solutions ~in the form of an oscillatory function times a damping term (i.e., a decaying exponential) in the direction of the propagation of the EM wave.

A Complex plane wave type solution for E and B associated with such wave equations are of the general form.

$$\vec{\tilde{E}} = \overrightarrow{\tilde{E}o}\, e^{i(\tilde{K}\, z - \omega\, t)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (9)$$

$$\vec{\tilde{B}} = \overrightarrow{\tilde{B}o}\, e^{i(\tilde{K}\, z - \quad t)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots.. (10)$$

With complex wave number

$\tilde{K} = \mathrm{k} + i\mathrm{K}$ ……………….. (11)

$$\tilde{K} = \tilde{K}(\omega)$$

From Eqs. 9, 10, 7 & 8

$\tilde{K}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega$ ………………….. (12)

Then,

$\tilde{K}^2 = (\mathrm{k} + i\mathrm{K})^2 = \mathrm{k}^2 - \mathrm{K}^2 + 2i\mathrm{kK} = \mu\epsilon\omega^2 + i\mu\sigma\omega$

Or $[(\mathrm{k}^2 - \mathrm{K}^2) - \mu\epsilon\omega^2] + i[2\mathrm{kK} - \mu\sigma\omega] = 0$ ……………….. (13)

Only way this relation can be true, when

$(\mathrm{k}^2 - \mathrm{K}^2) - \mu\epsilon\omega^2 = 0$

$2\mathrm{kK} - \mu\sigma\omega = 0$

$(\mathrm{k}^2 - \mathrm{K}^2) = \mu\epsilon\omega^2$ ………………. (14)

$2\mathrm{kK} = \mu\sigma\omega$ …………………….. (15)

We need to solve equations (14) and (15) simultaneously to determine k and Ҡ.

From equation (15), we get

$Ҡ = \frac{\mu\sigma\omega}{2k}$ ……………….. (16)

Plugging this into equation (14)

$k^2 – Ҡ^2 = k^2 - (\frac{\mu\sigma\omega}{2k})^2 = \mu\epsilon\omega^2$

Multiplying above equation by k^2, we get

$k^4 - (\frac{\mu\sigma\omega}{2})^2 = (\mu\epsilon\omega^2)k^2$

$k^4 - (\mu\epsilon\omega^2)k^2 - \left(\frac{1}{2}\mu\sigma\omega\right)^2 = 0$ ……………… (17)

Let's define $x = k^2$

$a = 1$

$b = -(\mu\epsilon\omega^2)$

$c = -(\frac{\mu\sigma\omega}{2})^2$

Then equation (17) becomes

$ax^2+bx+c = 0$

therefore, $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$

or, $k^2 = \frac{1}{2}[(\mu\epsilon\omega^2 \pm \sqrt{(\mu\epsilon\omega^2)^2+4(\frac{\mu\sigma\omega}{2})^2}$

or, $k^2 = \frac{1}{2}(\mu\epsilon\omega^2)[1 \pm \sqrt{1+4\frac{\mu^2\sigma^2\omega^2}{4\mu^2\epsilon^2\omega^4}}$]`

or, $k^2 = \frac{1}{2}(\mu\epsilon\omega^2)[1 \pm \sqrt{1+\frac{\sigma^2}{\epsilon^2\omega^2}}\ ]$

or, $k^2 = \frac{1}{2}(\mu\epsilon\omega^2)\left[1 \pm \sqrt{1+(\frac{\sigma}{\epsilon\omega})^2}\right]$ ………………. (18)

on physical grounds,($k^2>0$), we must select the +ve sign .

hence,

$$k^2 = \frac{1}{2}\,(\mu\epsilon\omega^2)\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right] \ldots\ldots\ldots\ldots\ldots \quad (19)$$

$$\text{or } k=\sqrt{k^2} = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\,\right]}$$

$$\text{or } k= \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\,\right]} \quad \ldots\ldots\ldots\ldots\ldots \; (20)$$

having thus solved for k , we can use either of our original two relations to solve for Ҡ . for e.g. $k^2 - Ҡ^2 = \mu\epsilon\omega^2$,

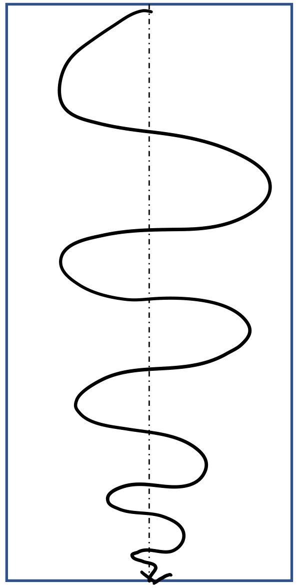
$$Ҡ^2 = k^2 - \mu\epsilon\omega^2 = \frac{1}{2}\,(\mu\epsilon\omega^2)\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right] - \mu\epsilon\omega^2$$

$$\text{Or, } Ҡ^2 = \frac{1}{2}\,(\mu\epsilon\omega^2)\left[-1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right] \ldots\ldots\ldots\ldots\ldots \; (21)$$

We thus have

$$k= \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\,\right]}$$

$$Ҡ = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[-1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\,\right]}$$



The imaginary part of $\widetilde{K}$ results in an attenuation of the wave (decreasing amplitude with increasing z)

$$\vec{\tilde{E}}(z,t) = \vec{\tilde{E}}_0 e^{-\mathrm{K}z}\, e^{i(k\,z-\omega t\,)} \qquad \ldots\ldots\ldots\ldots (23)$$

$$\vec{\tilde{B}}(z,t) = \overrightarrow{\tilde{B}_0} e^{-\mathrm{K}z}\, e^{i(k\,z-\omega t\,)}$$

Skin depth: The distance it takes to reduce the amplitude by a factor of $^1/_e$

$$d = \frac{1}{\mathrm{K}} \quad \ldots\ldots\ldots\ldots.. (24)$$ where d is the skin depth.

It is a measure of how far the wave penetrates into the conductor. Meanwhile the real part of $\widetilde{K}$ determines the wavelength the propagation speed, and the index of reflection,

$$\lambda = \frac{2\pi}{k} \quad \ldots\ldots\ldots. (25), \quad v=\frac{\omega}{k} \quad \ldots\ldots\ldots\ldots. (26), \quad n = \frac{ck}{\omega} \quad \ldots\ldots\ldots\ldots (27)$$

Maxwell's equation (6) impose further constraints fixes relatives amplitudes, phases and polarizations of E and B.

Eq. 6(i) and (ii) rule out any z components: the fields are transverse.

Assuming E is polarized along the x- direction:

$$\tilde{\vec{E}}(z,t) = \tilde{E}o\; e^{-\kappa z}\, e^{i(k z-\omega t)}\, \hat{x} \;\ldots\ldots\ldots\ldots\ldots\ (28)$$

Then equation 6(iii) gives,

$$\tilde{\vec{B}}(z,t) = \frac{\tilde{K}}{\omega}\tilde{E}o\; e^{-\kappa z}\, e^{i(k z-\omega t)}\, \hat{y} \;\ldots\ldots\ldots\ (29)$$

Since $B_y = \frac{k}{\omega} E_x$

$\Rightarrow$ E and B Fields are mutually perpendicular.

where $K = |\tilde{K}| = \sqrt{\kappa^2 + k^2} = \omega\sqrt{\mu\epsilon\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}}$ ………….. (30)

And $\emptyset = \tan^{-1}\frac{\kappa}{k}$ ……… (31)

$\tilde{K}$ can be expressed in terms of its modulus and phase:

$$\tilde{K} = k\, e^{i\emptyset}$$

According to equation (28) and (29), the complex amplitude

$\tilde{E}o = Eo\; e^{i\delta_E}$ and $\tilde{B}o = Bo e^{i\delta_B}$ (free space case) are related here by

$$Bo e^{i\delta_B} = \frac{k e^{i\varphi}}{\omega} Eo\; e^{i\delta_E} \;\ldots\ldots\ (32)$$

Thus, it is evident that the electric and magnetic fields are no longer in phase. In fact,

$\delta_B - \delta_E = \emptyset$ implies that magnetic field lags behind the electric field.

The real amplitudes of $\vec{E}$ and $\vec{B}$ are related by

$$\frac{Eo}{Bo} = \frac{k}{\omega} = \sqrt{\mu\epsilon\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}} \;\ldots\ldots\ldots\ldots\ldots\ (34)$$

Thanks